dijkstra's algorithm

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an algorithm that finds the distances from a given vertex to all the other vertices in a weighted graph

Input:

1. a graph $G = (V, E, w)$, the graph G is weighted non-negatively ($w(e) \geq 0, e \in E$)
2. a vertex $s \in V$ (the origin)

Output: for every vertex $v \in V$ two variables, $d[v], p[v]$ with the following restrictions

1. $d[v] = \delta(s, v)$
2. $p[v]$ is the second-to-last vertex in the minimal path from s to v

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DIJKSTRA(G,s):
    DIJ-INIT(G,s)
    DIJ-OP(G,s)
DIJ-INIT(G,s):
    d[s] ← 0
    p[s] ← Null
    for v ∈ V − {s} do
        d[v] ← ∞
        p[v] ← Null
DIJ-OP(G,s):
    PQ ← V  // priority queue
    while PQ ≠ ∅ do
        u ← PQ.extractMin()  // vertex with minimal d value
        for v ∈ adj[u] do
            RELAX(G,u,v)
    return d, p
RELAX(G,u,v):
    if d[v] > d[u] + w(u, v) then
        d[v] ← d[u] + w(u, v)
        p[v] ← u
```

Example 0.0.1. given a weighted graph $G = (V, E, w)$ that represents a town's junctions/roads
given $L \subseteq V$ is a subset of junctions with streetlights
given $s \in V$, propose an algorithm that given G, L, s would find the length of the shortest paths from s to each of the vertices in the graph, such that each path wouldnt contain more than 2 junctions with streetlights

Answer 0.0.1.

```
V' ← V × {0,1,2}
E' ← {((u,0),(v,0)),((u,1),(v,1)),((u,2),(v,2)) | (u,v) ∈ E, u ∉
    L} ∪ {((u,0),(v,1)),((u,1),(v,2)) | (u,v) ∈ E, u ∈ L}
w'((u,x),(v,y)) = w(u,v)    ((u,x),(v,y)) ∈ E'
DIJKSTRA(G', (s,0))
return min{d((v,0)), d((v,1)), d((v,2))} for each v ∈ V
```

complexity of this algorithm is $\Theta(|V|^2)$